

Market Sentinel

Complete Server Deployment Guide
NSE Options IV-Accumulation Trading Daemon

Stack	FastAPI + asyncio (Python 3.11) · React 18 + TypeScript
Python	3.11.x
Node	22.x (for frontend build only)
OS	Ubuntu 22.04 LTS (AWS Lightsail / any VPS)
Web server	Nginx (reverse proxy) + uvicorn (ASGI)
Database	SQLite viaaiosqlite (data/market_sentinel.db)
Market data	Dhan HQ WebSocket + REST API

1. Server Prerequisites

Start from a fresh **Ubuntu 22.04 LTS** instance (minimum 2 vCPU / 2 GB RAM). All commands below are run as the **ubuntu** user unless noted.

1.1 Update & install system dependencies

```
sudo apt-get update && sudo apt-get upgrade -y
sudo apt-get install -y \
    python3.11 python3.11-venv python3-pip \
    nginx git curl unzip build-essential
```

1.2 Install Node 22 (LTS)

```
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt-get install -y nodejs
node --version    # should print v22.x.x
npm --version
```

1.3 Verify Python 3.11

```
python3.11 --version    # should print Python 3.11.x
```

2. Clone & Configure the Repository

2.1 Create application directory and clone

```
sudo mkdir -p /opt/market-sentinel
sudo chown ubuntu:ubuntu /opt/market-sentinel

# Option A – clone from GitHub (if repo is on GitHub)
git clone https://github.com/<your-org>/market-sentinel.git /opt/market-sentinel

# Option B – copy from local machine via scp
scp -r /path/to/market-sentinel ubuntu@<SERVER_IP>:/opt/market-sentinel
```

2.2 Create the environment file

The app reads credentials from **/opt/market-sentinel/.env** at startup.

```
cp /opt/market-sentinel/.env.example /opt/market-sentinel/.env
nano /opt/market-sentinel/.env
```

.env contents:

```
DHAN_CLIENT_ID=your_dhan_client_id_here
DHAN_ACCESS_TOKEN=your_dhan_access_token_here
```

Get your Client ID and Access Token from the Dhan portal → My Apps.

2.3 Review config.yaml

All tuning parameters live in **config.yaml**. Key settings to verify:

```
# /opt/market-sentinel/config.yaml (excerpt)

dhan:
  client_id: "your_client_id"      # also set in .env
  access_token: "${DHAN_ACCESS_TOKEN}" # loaded from .env

execution:
  paper_trading: true              # ← change to false for live trading
  virtual_capital: 100000          # ■1 lakh virtual capital

ws:
  warmup_seconds: 300              # 5-min blackout after WS reconnect
```

Keep paper_trading: true until you are confident the system is working correctly.

3. Python Virtual Environment

3.1 Create venv and install dependencies

```
cd /opt/market-sentinel
python3.11 -m venv .venv
.venv/bin/pip install --upgrade pip
.venv/bin/pip install -r requirements.txt
```

3.2 Python packages installed

Package	Purpose
fastapi==0.115.5	ASGI web framework
uvicorn[standard]==0.32.1	ASGI server
websockets==13.1	Dhan WS binary feed
aiosqlite==0.20.0	Async SQLite wrapper
pyyaml==6.0.2	config.yaml parser
numpy==2.1.3	Rolling statistics
scipy==1.14.1	Z-score & delta calculations
python-dotenv==1.0.1	.env file loader
httpx==0.27.2	Async HTTP client (PCR polling)
dhanhq>=2.0.0	Dhan HQ SDK (scrip master)

3.3 Verify backend starts

```
cd /opt/market-sentinel
.venv/bin/uvicorn api.main:app --host 0.0.0.0 --port 8000
# Press Ctrl+C after confirming "Application startup complete."
```

4. Frontend Build (React)

4.1 Install npm dependencies and build

```
cd /opt/market-sentinel/frontend
npm install
npm run build
```

This creates **/opt/market-sentinel/frontend/dist/** — the static files nginx will serve.

4.2 Verify build

```
ls /opt/market-sentinel/frontend/dist/
# Expected: index.html  assets/  ...
```

5. Nginx Configuration

5.1 Install nginx config

```
sudo cp /opt/market-sentinel/nginx.conf /etc/nginx/sites-available/market-sentinel
sudo ln -sf /etc/nginx/sites-available/market-sentinel \
        /etc/nginx/sites-enabled/market-sentinel
sudo rm -f /etc/nginx/sites-enabled/default # remove default site
sudo nginx -t                               # test config
sudo systemctl reload nginx
```

5.2 nginx.conf (full file)

```
# /etc/nginx/sites-available/market-sentinel
server {
    listen 80;
    server_name _;    # replace with your domain or static IP

    # React static assets
    root /opt/market-sentinel/frontend/dist;
    index index.html;

    # SPA fallback
    location / {
        try_files $uri $uri/ /index.html;
    }

    # API proxy
    location /api/ {
        proxy_pass            http://127.0.0.1:8000;
        proxy_set_header     Host $host;
        proxy_set_header     X-Real-IP $remote_addr;
        proxy_set_header     X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_read_timeout   60s;
    }

    # WebSocket proxy
    location /ws/ {
        proxy_pass            http://127.0.0.1:8000;
        proxy_http_version   1.1;
        proxy_set_header     Upgrade $http_upgrade;
        proxy_set_header     Connection "upgrade";
        proxy_set_header     Host $host;
        proxy_read_timeout   3600s;
    }

    # Health check
    location /health {
        proxy_pass http://127.0.0.1:8000/health;
    }
}
```

If you have a domain name, replace server_name _; with server_name yourdomain.com;

6. Systemd Service

6.1 Install and enable the service

```
sudo cp /opt/market-sentinel/market-sentinel.service /etc/systemd/system/  
sudo systemctl daemon-reload  
sudo systemctl enable market-sentinel    # start on boot
```

6.2 market-sentinel.service (full file)

```
[Unit]  
Description=Market Sentinel – NSE options IV-accumulation trading daemon  
After=network-online.target  
Wants=network-online.target  
  
[Service]  
Type=simple  
User=ubuntu  
WorkingDirectory=/opt/market-sentinel  
EnvironmentFile=/opt/market-sentinel/.env  
ExecStart=/opt/market-sentinel/.venv/bin/uvicorn api.main:app \  
    --host 0.0.0.0 --port 8000 \  
    --workers 1 \  
    --log-level info  
Restart=on-failure  
RestartSec=10  
TimeoutStopSec=30  
NoNewPrivileges=true  
PrivateTmp=true  
  
[Install]  
WantedBy=multi-user.target
```

6.3 Start & verify

```
sudo systemctl start market-sentinel  
sudo systemctl status market-sentinel    # should show "active (running)"  
sudo journalctl -u market-sentinel -f    # tail live logs
```

7. Data Directories & Permissions

```
mkdir -p /opt/market-sentinel/data/ticks
# SQLite DB is auto-created at first startup: data/market_sentinel.db
# Tick CSVs are written to: data/ticks/YYYY-MM-DD.csv
```

7.1 Optional: daily tick CSV purge (cron)

Tick CSV files accumulate — add a midnight cron to purge files older than 1 day:

```
crontab -e
# Add this line:
0 0 * * * find /opt/market-sentinel/data/ticks -name '*.csv' -mtime +1 -delete
```

8. One-Shot Setup Script

The repo includes **scripts/setup.sh** which runs all steps 3–7 automatically.

```
cd /opt/market-sentinel
chmod +x scripts/setup.sh
./scripts/setup.sh
```

Then finish manually:

```
cp /opt/market-sentinel/.env.example /opt/market-sentinel/.env
nano /opt/market-sentinel/.env          # paste your Dhan credentials
sudo systemctl start market-sentinel
```

9. Verification Checklist

Check	Command / Expected output
Backend health	<code>curl http://localhost:8000/health</code> → <code>{"status":"ok",...}</code>
Frontend	Open <code>http://<SERVER_IP>/</code> in browser – dashboard loads
API via nginx	<code>curl http://<SERVER_IP>/api/status</code> → JSON with engine state
WebSocket	Browser DevTools → Network → WS → <code>/ws/feed</code> shows live frames
Service auto-start	<code>sudo reboot; sudo systemctl status market-sentinel</code>
Logs	<code>sudo journalctl -u market-sentinel -n 100 --no-pager</code>

10. Day-to-Day Operations

10.1 Service control

```
sudo systemctl start    market-sentinel # start
sudo systemctl stop     market-sentinel # graceful stop (30 s timeout)
sudo systemctl restart  market-sentinel # restart
sudo systemctl status   market-sentinel # status + recent logs
```

10.2 Live log tailing

```
sudo journalctl -u market-sentinel -f          # follow
sudo journalctl -u market-sentinel -n 200       # last 200 lines
sudo journalctl -u market-sentinel --since today # today only
```

10.3 Update Dhan Access Token

Dhan tokens expire daily. Update without restarting the service:

```
# Option A – API endpoint (no restart needed):
curl -X POST http://localhost:8000/api/token \
  -H "Content-Type: application/json" \
  -d '{"token": "NEW_ACCESS_TOKEN_HERE"}'

# Option B – update .env and restart:
nano /opt/market-sentinel/.env      # change DHAN_ACCESS_TOKEN=...
sudo systemctl restart market-sentinel
```

10.4 Deploy code update

```
cd /opt/market-sentinel
git pull origin main

# Reinstall Python deps if requirements.txt changed:
.venv/bin/pip install -r requirements.txt

# Rebuild frontend if frontend/ changed:
cd frontend && npm install && npm run build && cd ..

sudo systemctl restart market-sentinel
```

10.5 Clear trading records (paper trades / signals)

```
curl -X POST http://localhost:8000/api/records/clear \
  -H "Content-Type: application/json" \
  -d '{"trades": true, "signals": true}'
```

10.6 Toggle mock mode (no live WS feed needed)

```
# In config.yaml: (no such field – use paper_trading instead)
execution:
  paper_trading: true    # true = paper trade, false = live order placement
```

11. Firewall & Security

11.1 UFW (recommended)

```
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'    # ports 80 and 443
sudo ufw deny 8000/tcp        # block direct uvicorn access from outside
sudo ufw enable
sudo ufw status
```

11.2 Dhan portal: whitelist server IP

Log in to **dhan.co** → **My Apps** → **your app** → **Allowed IPs** and add your server's public IP. Without this, Dhan WS returns code=1006 (connection refused after handshake).

11.3 SSL / HTTPS (optional but recommended)

```
sudo apt-get install -y certbot python3-certbot-nginx
sudo certbot --nginx -d yourdomain.com
# Certbot auto-renews; verify: sudo certbot renew --dry-run
```

12. Troubleshooting Quick Reference

Symptom	Fix
WS code=1006	Server IP not whitelisted in Dhan portal → My Apps → Allowed IPs
Auth failure / 10-min backoff	Dhan Access Token expired — update via POST /api/token
PCR always WAIT	PCR poll returning 429 — sequential mode is default; check Dhan subscription
No ticks in feed	Market closed, or Dhan Market Feed subscription not active
Frontend shows stale ATM	Refresh browser; WebSocket heartbeat sends instrument_map every 5s
uvicorn port 8000 in use	sudo lsof -i :8000 → kill the old process
permission denied on .db	sudo chown ubuntu:ubuntu /opt/market-sentinel/data/ -R
npm build fails	Node version mismatch — run node --version, must be v22.x

13. Complete Quick-Start Command Sequence

Copy-paste in order on a fresh Ubuntu 22.04 server:

```
# 1. System packages
sudo apt-get update && sudo apt-get upgrade -y
sudo apt-get install -y python3.11 python3.11-venv python3-pip nginx git curl build-essential

# 2. Node 22
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt-get install -y nodejs

# 3. Clone repo (replace URL)
sudo mkdir -p /opt/market-sentinel
sudo chown ubuntu:ubuntu /opt/market-sentinel
git clone https://github.com/<your-org>/market-sentinel.git /opt/market-sentinel

# 4. Credentials
cp /opt/market-sentinel/.env.example /opt/market-sentinel/.env
nano /opt/market-sentinel/.env          # fill DHAN_CLIENT_ID, DHAN_ACCESS_TOKEN

# 5. Python venv
cd /opt/market-sentinel
python3.11 -m venv .venv
.venv/bin/pip install --upgrade pip
.venv/bin/pip install -r requirements.txt

# 6. Frontend build
cd /opt/market-sentinel/frontend
npm install && npm run build
cd /opt/market-sentinel

# 7. Data directory
mkdir -p data/ticks

# 8. Nginx
sudo cp nginx.conf /etc/nginx/sites-available/market-sentinel
sudo ln -sf /etc/nginx/sites-available/market-sentinel /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t && sudo systemctl reload nginx

# 9. Systemd service
sudo cp market-sentinel.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable market-sentinel
sudo systemctl start market-sentinel

# 10. Verify
sudo systemctl status market-sentinel
curl http://localhost:8000/health
# Then open http://<SERVER_IP>/ in browser
```